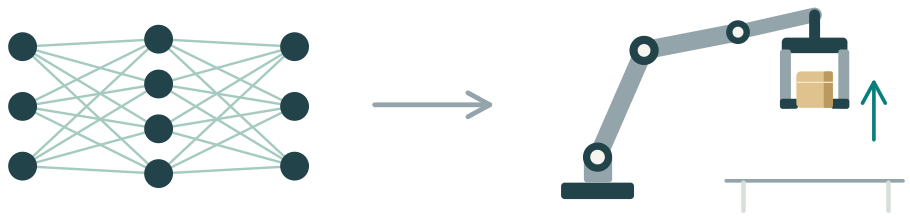
Adaptation needs capability and information

How can a robot execute a task in a situation it did not encounter during training?

1 Movement capability

Learn how to grasp and move
across variations of the same task.



2 Environment interpretation

Estimate the current conditions
so the policy can use suitable actions.



Working hypothesis: generalizable behavior and useful environmental context support adaptation.

An RL policy conditioned on environment

Standard state describes the robot and task; environmental parameters describe current conditions.

Robot + task state

Joints, gripper pose, task state

Environmental parameters

Known in training;
estimated at deployment



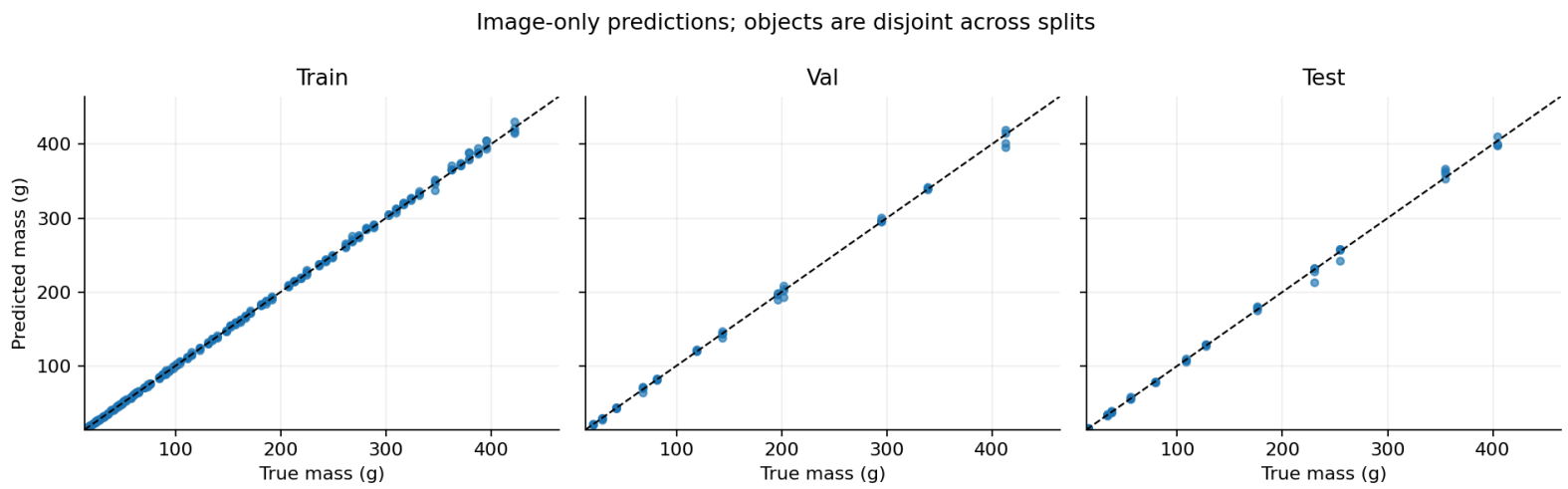
Grasp and move actions

Training: learn policy weights from reward. Deployment test: keep those weights fixed.

Assumption to test: varied training produces a policy capable of executing at held-out heights.

A controlled visual mass estimate works

Fixed-density cubes provide a visible size-to-mass relationship.



TEST MASS ERROR

2.76 g

Mean absolute error

Constant baseline

106.37 g

12 held-out sizes / 48 test images

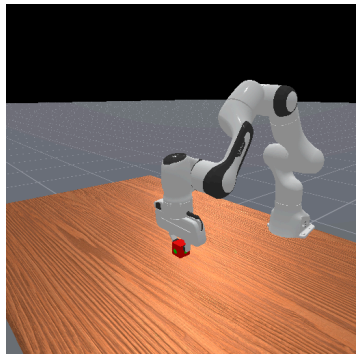
Scope: one material, fixed density and friction, fixed camera. This tests interpolation over size.

Evidence: runs/visual_mass/initial_rgb_mass/model/metrics.json and predictions.png

The full pipeline runs in simulation

Initial RGB image → mass estimate → supplied policy input → pick-and-lift execution.

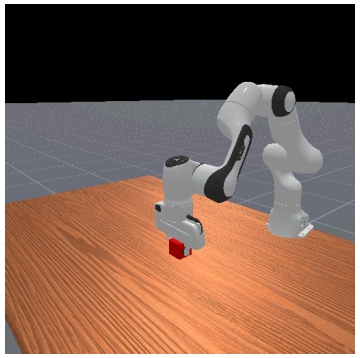
4 cm cube



64.0 g → 62.32 g

actual mass → predicted mass

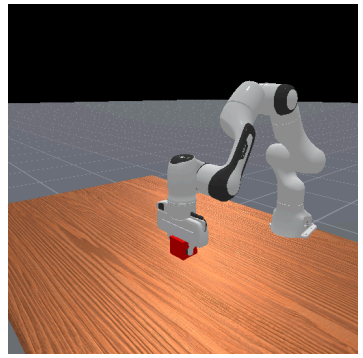
5 cm cube



125.0 g → 121.24 g

actual mass → predicted mass

6 cm cube



216.0 g → 212.45 g

actual mass → predicted mass

3 configured trials · 3 final successes · 50 actions per trial

One seed per size. Integration is demonstrated; estimation benefit and unseen-height adaptation remain untested.

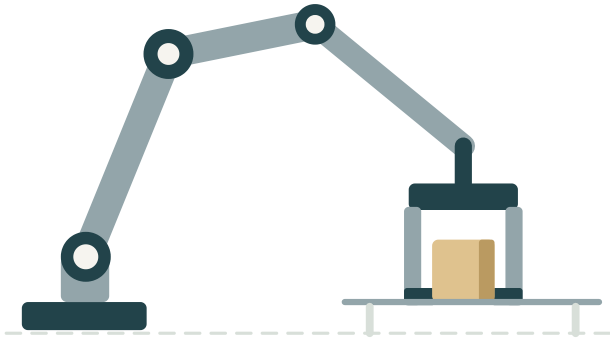
Evidence: runs/deployment/20261022_014201_067220/summary.json, manifest.json, and final frames

TO BE DONE

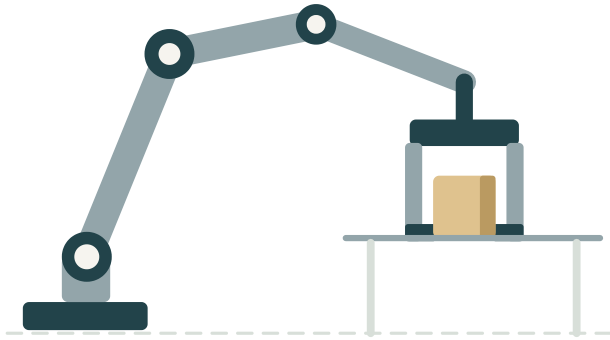
Train pick and lift across object heights

Vary initial object elevation relative to the arm while keeping the task and grasp strategy consistent.

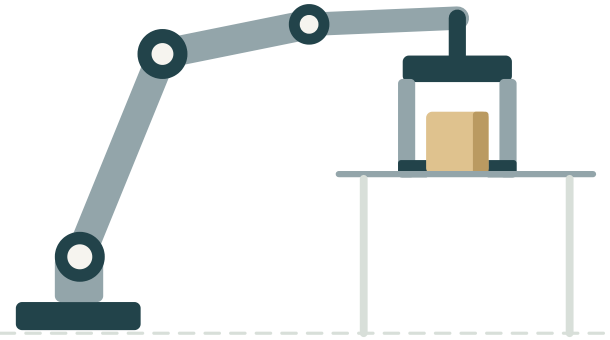
Training height h1



Training height h2



Training height h3

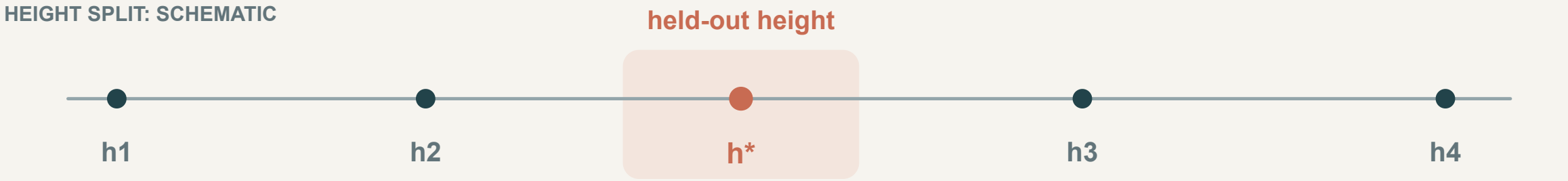


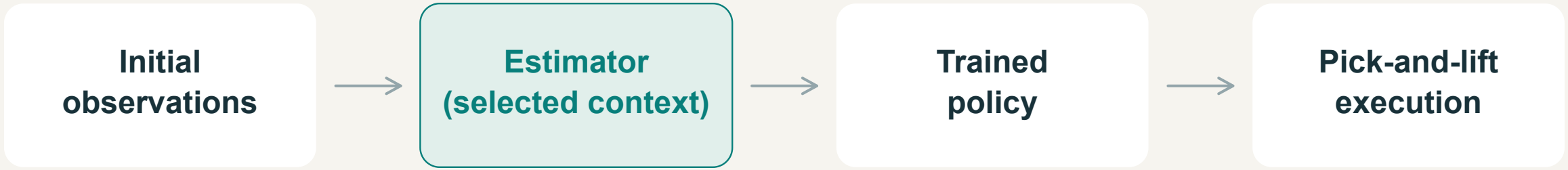
The robot base stays fixed. The object starts at different elevations.

First height study: control mass, friction, object geometry, and required lift clearance.

Deploy at a held-out initial height

Freeze the trained policy and provide environmental context estimated from deployment observations.





Report within-range held-out heights separately from heights beyond the training range.

Current estimator: mass. Estimator target for the height study still needs to be specified.

Separate policy capability from estimation

At each held-out height, compare the same frozen policy under three context inputs.

CONTEXT INPUT	QUESTION
True context	Can the policy execute at the unseen height?
Estimated context	How much performance is lost to estimation?
Fixed / incorrect context	Does useful context improve execution?

1. Reserve test heights before training.
2. Match physical scenes and starts.
3. Freeze models; predefine held success.
4. Remove true copies of estimated inputs.

What we can say now—and test next

Current evidence supports the components; the proposed comparison must establish adaptation.

DEMONSTRATED

Controlled image-to-mass estimate

2.76 g MAE on held-out cube sizes

Estimator-to-policy integration

Three configured full-horizon successes

NEXT EXPERIMENT

Train across initial object heights

Establish policy capability with true context

Evaluate estimated context at held-out heights

Compare against true and fixed inputs

If true context fails: inspect policy capability and task feasibility.

If true context succeeds but estimates fail: inspect estimation and policy sensitivity.

Later stage: revise environmental estimates through interaction, success, and failure.